Class 08: Breast Cancer Mini Project

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Background

The goal of this mini-project is for you to explore a complete analysis using the unsupervised learning techniques covered in class.

Today we will analyze a biopsy data-set from fine needle aspiration (FNA) of a breast mass.

Data Import

The data is made available as a CSV file for download. We can read this in using `read.csv()`:

```
fna.data <- "WisconsinCancer.csv"
```

```
wisc.df <- read.csv(fna.data, row.names = 1)
```

```r
head(wisc.df, 4)
```

```
         diagnosis radius_mean texture_mean perimeter_mean area_mean
842302           M       17.99        10.38         122.80    1001.0
842517           M       20.57        17.77         132.90    1326.0
84300903         M       19.69        21.25         130.00    1203.0
84348301         M       11.42        20.38          77.58     386.1
         smoothness_mean compactness_mean concavity_mean concave.points_mean
842302           0.11840          0.27760         0.3001             0.14710
842517           0.08474          0.07864         0.0869             0.07017
84300903         0.10960          0.15990         0.1974             0.12790
84348301         0.14250          0.28390         0.2414             0.10520
         symmetry_mean fractal_dimension_mean radius_se texture_se perimeter_se
842302          0.2419                0.07871    1.0950     0.9053        8.589
842517          0.1812                0.05667    0.5435     0.7339        3.398
84300903        0.2069                0.05999    0.7456     0.7869        4.585
84348301        0.2597                0.09744    0.4956     1.1560        3.445
         area_se smoothness_se compactness_se concavity_se concave.points_se
842302    153.40      0.006399        0.04904      0.05373           0.01587
842517     74.08      0.005225        0.01308      0.01860           0.01340
84300903   94.03      0.006150        0.04006      0.03832           0.02058
84348301   27.23      0.009110        0.07458      0.05661           0.01867
         symmetry_se fractal_dimension_se radius_worst texture_worst
842302       0.03003             0.006193        25.38         17.33
842517       0.01389             0.003532        24.99         23.41
84300903     0.02250             0.004571        23.57         25.53
84348301     0.05963             0.009208        14.91         26.50
         perimeter_worst area_worst smoothness_worst compactness_worst
842302            184.60     2019.0           0.1622            0.6656
842517            158.80     1956.0           0.1238            0.1866
84300903          152.50     1709.0           0.1444            0.4245
84348301           98.87      567.7           0.2098            0.8663
         concavity_worst concave.points_worst symmetry_worst
842302            0.7119               0.2654         0.4601
842517            0.2416               0.1860         0.2750
84300903          0.4504               0.2430         0.3613
84348301          0.6869               0.2575         0.6638
         fractal_dimension_worst
842302                   0.11890
842517                   0.08902
84300903                 0.08758
84348301                 0.17300
```

Make sure we remove or exclude the `diagnosis` column from the data-set that we use for further analysis - this is the expert diagnosis as either M or B:

```
wisc.data <- wisc.df[, -1]
diagnosis <- as.factor( wisc.df$diagnosis )
```

Exploratory Data Analysis

> Q1. How many observations are in this dataset?

569 observations

```
nrow(wisc.data)
```

```
[1] 569
```

> Q2. How many of the observations have a malignant diagnosis?

212 malignant diagnosis

```
table(wisc.df$diagnosis)
```

```

  B   M 
357 212 
```

```
sum( wisc.df$diagnosis == "M")
```

```
[1] 212
```

> Q3. How many variables/features in the data are suffixed with `_mean`?

10 variables

We can use the `grep()` function to help us here:

```
length( grep("_mean$", colnames(wisc.df), value = TRUE) )
```

```
[1] 10
```

Principal Component Analysis (PCA)

We need to scale our data before PCA with the `scale=TRUE` argument to `prcomp()`

```r
colMeans(wisc.data)
```

```
            radius_mean            texture_mean          perimeter_mean 
           1.412729e+01            1.928965e+01            9.196903e+01 
              area_mean         smoothness_mean        compactness_mean 
           6.548891e+02            9.636028e-02            1.043410e-01 
         concavity_mean     concave.points_mean           symmetry_mean 
           8.879932e-02            4.891915e-02            1.811619e-01 
 fractal_dimension_mean               radius_se              texture_se 
           6.279761e-02            4.051721e-01            1.216853e+00 
           perimeter_se                 area_se           smoothness_se 
           2.866059e+00            4.033708e+01            7.040979e-03 
         compactness_se            concavity_se       concave.points_se 
           2.547814e-02            3.189372e-02            1.179614e-02 
            symmetry_se    fractal_dimension_se            radius_worst 
           2.054230e-02            3.794904e-03            1.626919e+01 
          texture_worst         perimeter_worst              area_worst 
           2.567722e+01            1.072612e+02            8.805831e+02 
       smoothness_worst       compactness_worst         concavity_worst 
           1.323686e-01            2.542650e-01            2.721885e-01 
   concave.points_worst          symmetry_worst fractal_dimension_worst 
           1.146062e-01            2.900756e-01            8.394582e-02 
```

```r
apply(wisc.data, 2, sd)
```

```
            radius_mean            texture_mean          perimeter_mean 
           3.524049e+00            4.301036e+00            2.429898e+01 
              area_mean         smoothness_mean        compactness_mean 
           3.519141e+02            1.406413e-02            5.281276e-02 
         concavity_mean     concave.points_mean           symmetry_mean 
           7.971981e-02            3.880284e-02            2.741428e-02 
 fractal_dimension_mean               radius_se              texture_se 
           7.060363e-03            2.773127e-01            5.516484e-01 
           perimeter_se                 area_se           smoothness_se 
           2.021855e+00            4.549101e+01            3.002518e-03 
         compactness_se            concavity_se       concave.points_se 
           1.790818e-02            3.018606e-02            6.170285e-03 
```

```
          symmetry_se    fractal_dimension_se            radius_worst
         8.266372e-03            2.646071e-03            4.833242e+00
        texture_worst         perimeter_worst              area_worst
         6.146258e+00            3.360254e+01            5.693570e+02
     smoothness_worst       compactness_worst         concavity_worst
         2.283243e-02            1.573365e-01            2.086243e-01
 concave.points_worst          symmetry_worst fractal_dimension_worst
         6.573234e-02            6.186747e-02            1.806127e-02
```

```r
wisc.pr <- prcomp(wisc.data, scale = TRUE)
summary(wisc.pr)
```

```
Importance of components:
                          PC1    PC2     PC3     PC4     PC5     PC6     PC7
Standard deviation     3.6444 2.3857 1.67867 1.40735 1.28403 1.09880 0.82172
Proportion of Variance 0.4427 0.1897 0.09393 0.06602 0.05496 0.04025 0.02251
Cumulative Proportion  0.4427 0.6324 0.72636 0.79239 0.84734 0.88759 0.91010
                           PC8    PC9    PC10   PC11    PC12    PC13    PC14
Standard deviation     0.69037 0.6457 0.59219 0.5421 0.51104 0.49128 0.39624
Proportion of Variance 0.01589 0.0139 0.01169 0.0098 0.00871 0.00805 0.00523
Cumulative Proportion  0.92598 0.9399 0.95157 0.9614 0.97007 0.97812 0.98335
                          PC15    PC16    PC17    PC18    PC19    PC20   PC21
Standard deviation     0.30681 0.28260 0.24372 0.22939 0.22244 0.17652 0.1731
Proportion of Variance 0.00314 0.00266 0.00198 0.00175 0.00165 0.00104 0.0010
Cumulative Proportion  0.98649 0.98915 0.99113 0.99288 0.99453 0.99557 0.9966
                          PC22    PC23   PC24    PC25    PC26    PC27    PC28
Standard deviation     0.16565 0.15602 0.1344 0.12442 0.09043 0.08307 0.03987
Proportion of Variance 0.00091 0.00081 0.0006 0.00052 0.00027 0.00023 0.00005
Cumulative Proportion  0.99749 0.99830 0.9989 0.99942 0.99969 0.99992 0.99997
                          PC29    PC30
Standard deviation     0.02736 0.01153
Proportion of Variance 0.00002 0.00000
Cumulative Proportion  1.00000 1.00000
```

> Q4. From your results, what proportion of the original variance is captured by the first principal component (PC1)?

PC1 captures about 44.3% of the original variance.

> Q5. How many principal components (PCs) are required to describe at least 70% of the original variance in the data?

3 PCs are required to describe at least 70% of the original variance.

> Q6. How many principal components (PCs) are required to describe at least 90% of the original variance in the data?

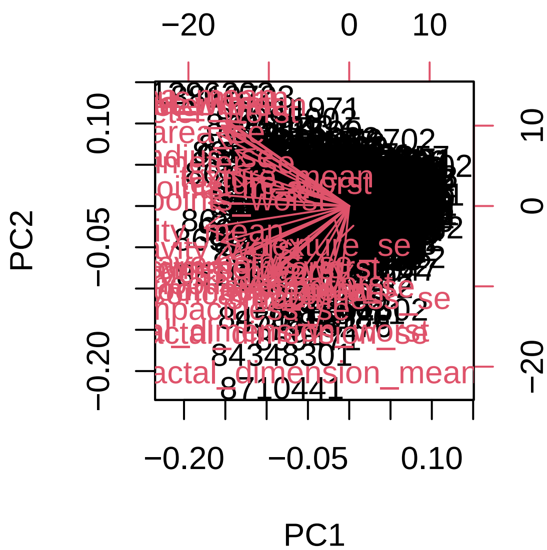
7 PCs are required to describe at least 90% of the original variance.

Interpreting PCA Results

Now you will use some visualizations to better understand your PCA model. A common visualization for PCA results is the so-called biplot.

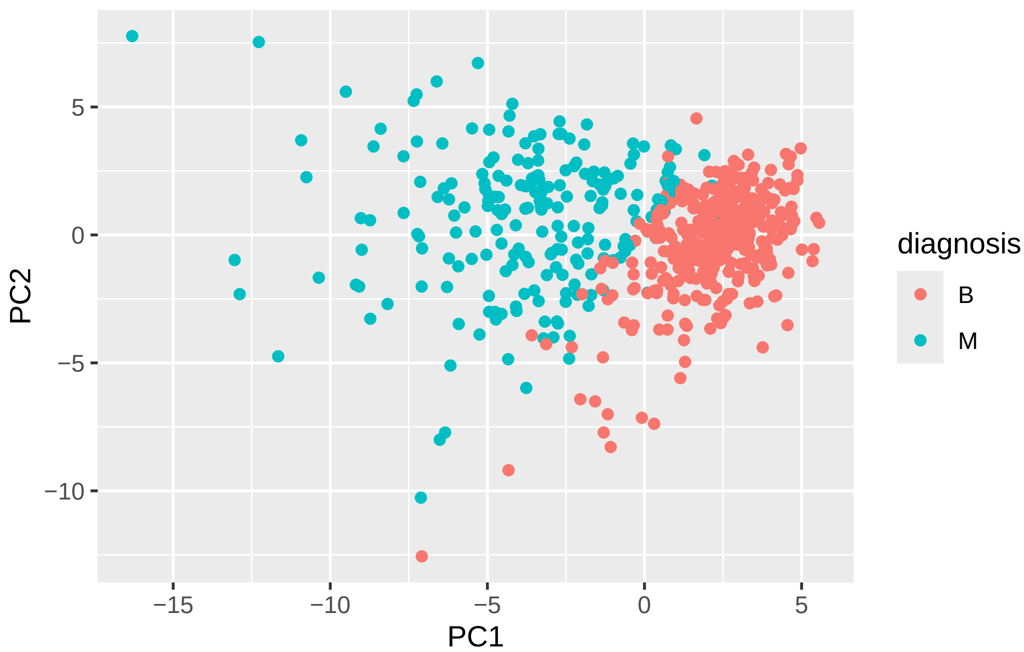
Create a biplot of the `wisc.pr` using the `biplot()` function.

```r
biplot(wisc.pr)
```



Let's see the "PC score plot"

```r
library(ggplot2)
ggplot(wisc.pr$x) +
  aes(PC1, PC2, col = diagnosis) +
  geom_point()
```



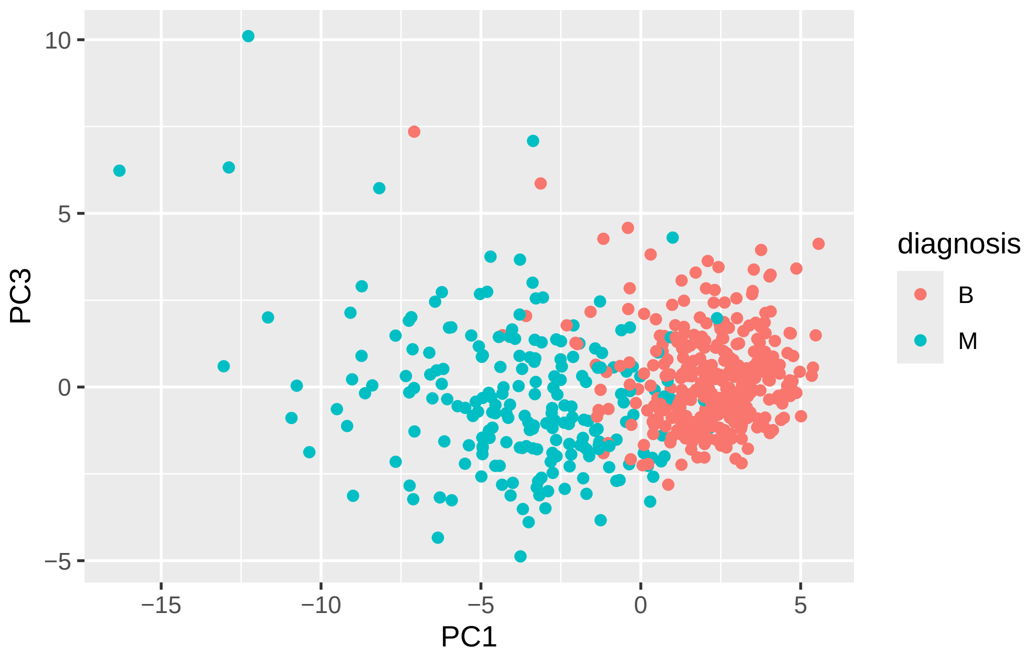
> Q7. What stands out to you about this plot? Is it easy or difficult to understand? Why?

The two clusters stood out to me on this plot. It is kind of difficult to understand because there are a lot of overlapping points in the plot, making it look crowded. It is also hard to know which original variables are driving the pattern.

> Q8. Generate a similar plot for principal components 1 and 3. What do you notice about these plots?

I noticed that there is still separation between benign and malignant samples, along PC1. PC1 vs PC3 is less clear compared to PC1 vs PC2 because PC3 has less variance.

```r
ggplot(wisc.pr$x) +
  aes(PC1, PC3, col = diagnosis) +
  geom_point()
```



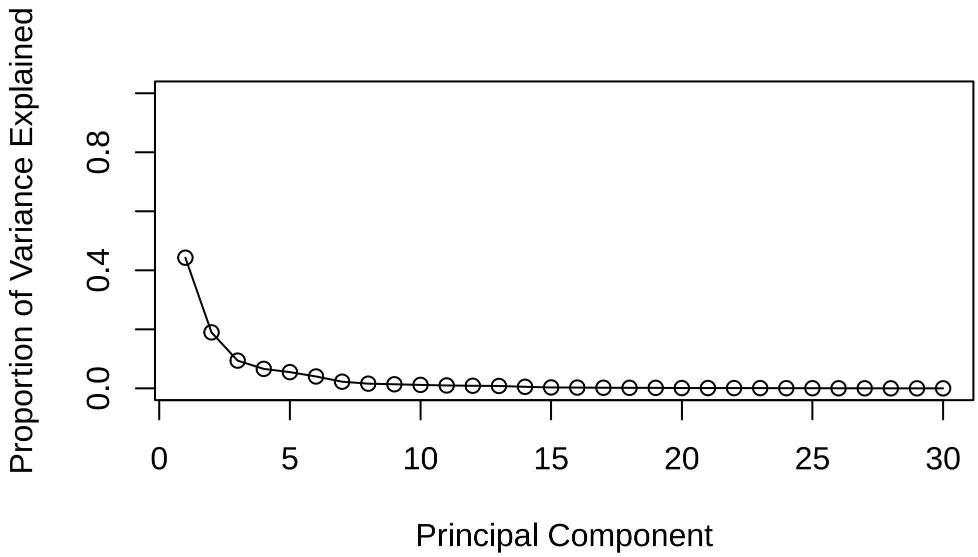
Variance Explained

```
# Calculate variance of each component
pr.var <- wisc.pr$sdev^2
head(pr.var)
```

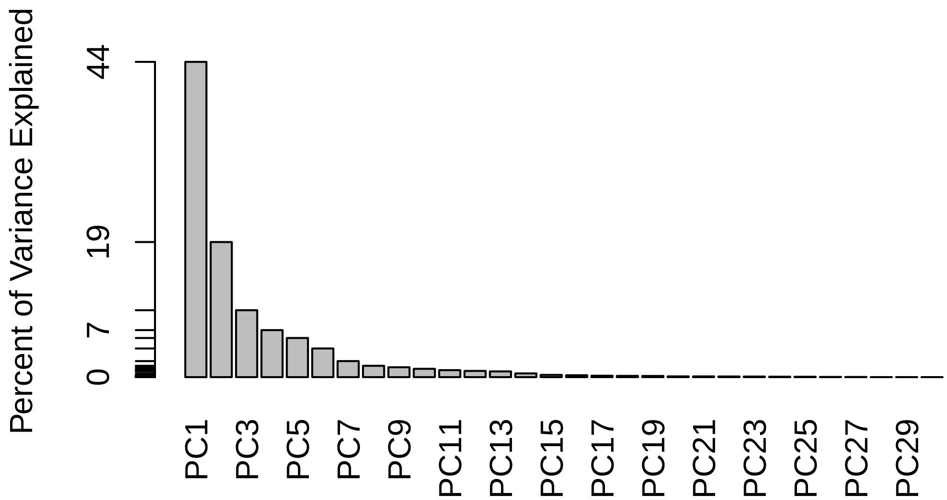
```
[1] 13.281608  5.691355  2.817949  1.980640  1.648731  1.207357
```

```
# Variance explained by each principal component
pve <- pr.var / sum(pr.var)

# Plot variance
plot(pve,
     xlab = "Principal Component",
     ylab = "Proportion of Variance Explained",
     ylim = c(0,1),
     type = "o")
```




```
#Alternative scree plot of same data, note data driven y-axis
barplot(pve, ylab = "Percent of Variance Explained",
     names.arg=paste0("PC",1:length(pve)), las=2, axes = FALSE)
axis(2, at=pve, labels=round(pve,2)*100 )
```



Communicating PCA Results

> Q9. For the first principal component, what is the component of the loading vector (i.e. `wisc.pr$rotation[,1]`) for the feature `concave.points_mean`? This tells us how much this original feature contributes to the first PC. Are there any features with larger contributions than this one?

The loading for `concave.points_mean` is -0.2608538. Yes, there are features with larger contributions than this one, such as `concavity_mean`.

```
wisc.pr$rotation["concave.points_mean", 1]
```

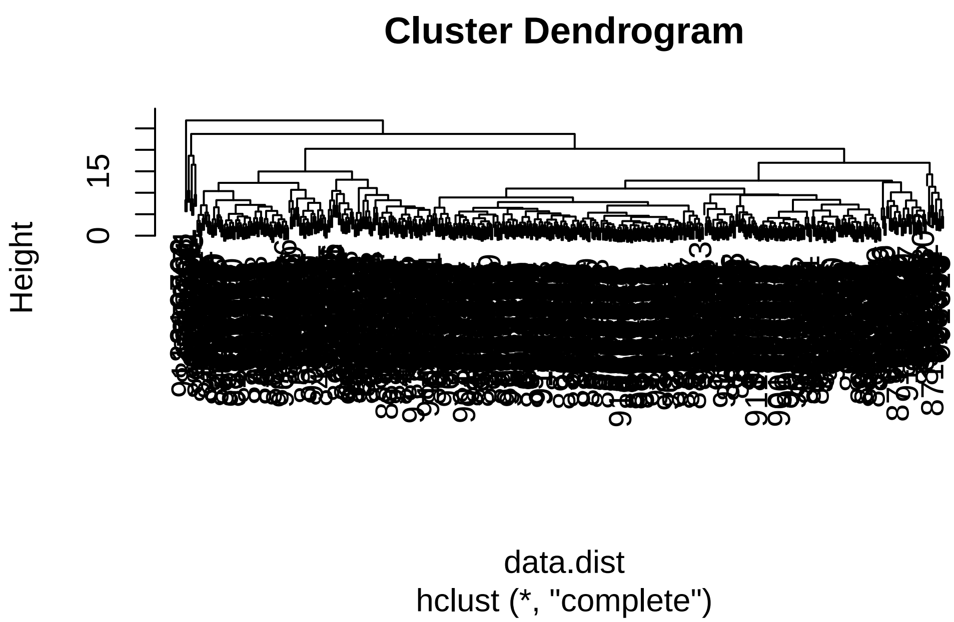
```
[1] -0.2608538
```

Hierarchical Clustering

```
data.scaled <- scale(wisc.data)
```

```r
data.dist <- dist(data.scaled)
```

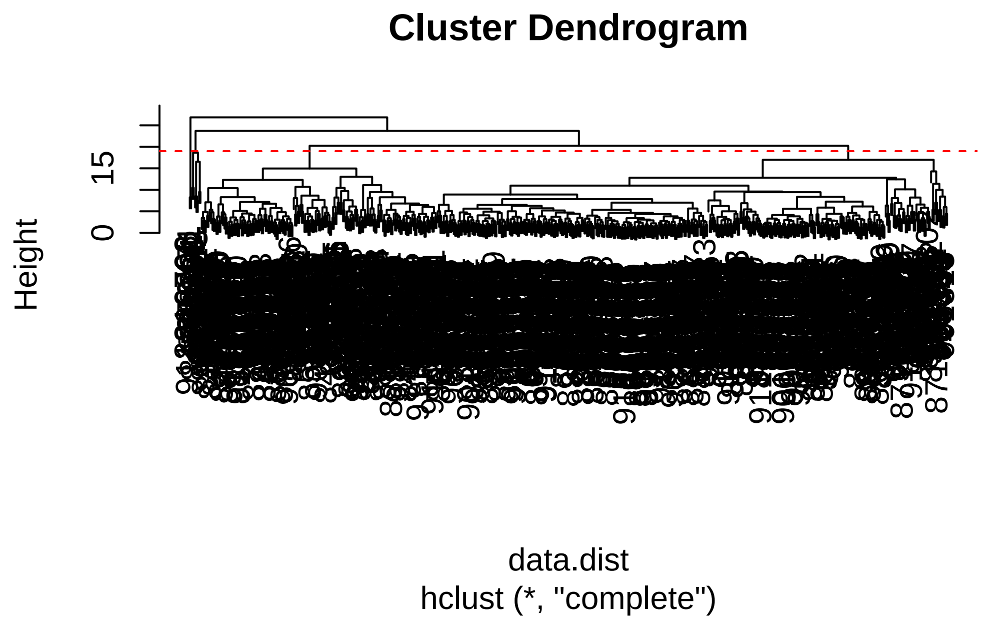
```r
wisc.hclust <- hclust(data.dist, method = "complete")
plot(wisc.hclust)
```



Results of Hierarchical Clustering

> Q10. Using the `plot()` and `abline()` functions, what is the height at which the clustering model has 4 clusters?

```r
plot(wisc.hclust)
abline(h = 19, col = "red", lty = 2)
```



Selecting Number of Clusters

```r
wisc.hclust.clusters <- cutree(wisc.hclust, k = 4)
```

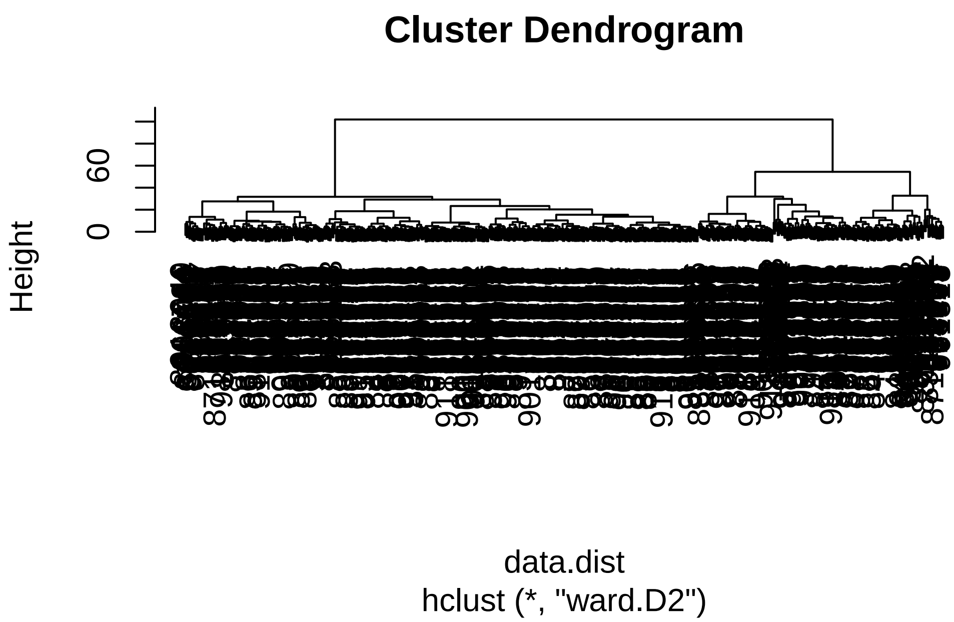
```r
table(wisc.hclust.clusters, diagnosis)
```

```
                    diagnosis
wisc.hclust.clusters   B   M
                   1  12 165
                   2   2   5
                   3 343  40
                   4   0   2
```

> Q12. Which method gives your favorite results for the same `data.dist` dataset? Explain your reasoning.

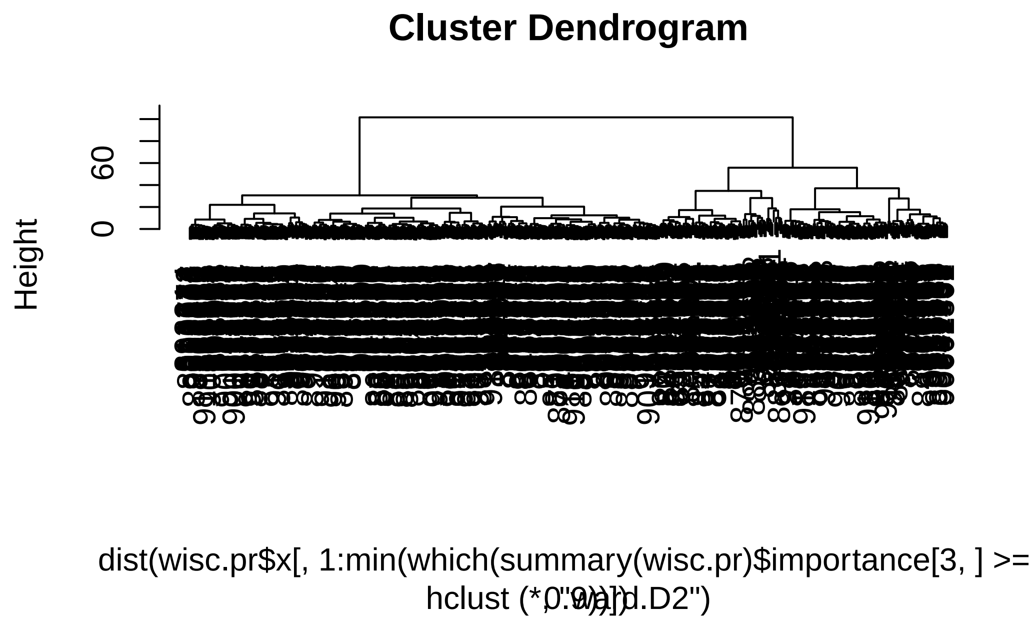
The ward.D2 method gives my favorite results for the dataset because it forms compact and well-separated clusters. It also minimizes the cluster variance.

```r
wisc.hclust.ward <- hclust(data.dist, method = "ward.D2")
plot(wisc.hclust.ward)
```



Combining Methods

```r
wisc.pr.hclust <- hclust(
  dist(wisc.pr$x[, 1:min(which(summary(wisc.pr)$importance[3, ] >= 0.90))]),
  method = "ward.D2")
plot(wisc.pr.hclust)
```



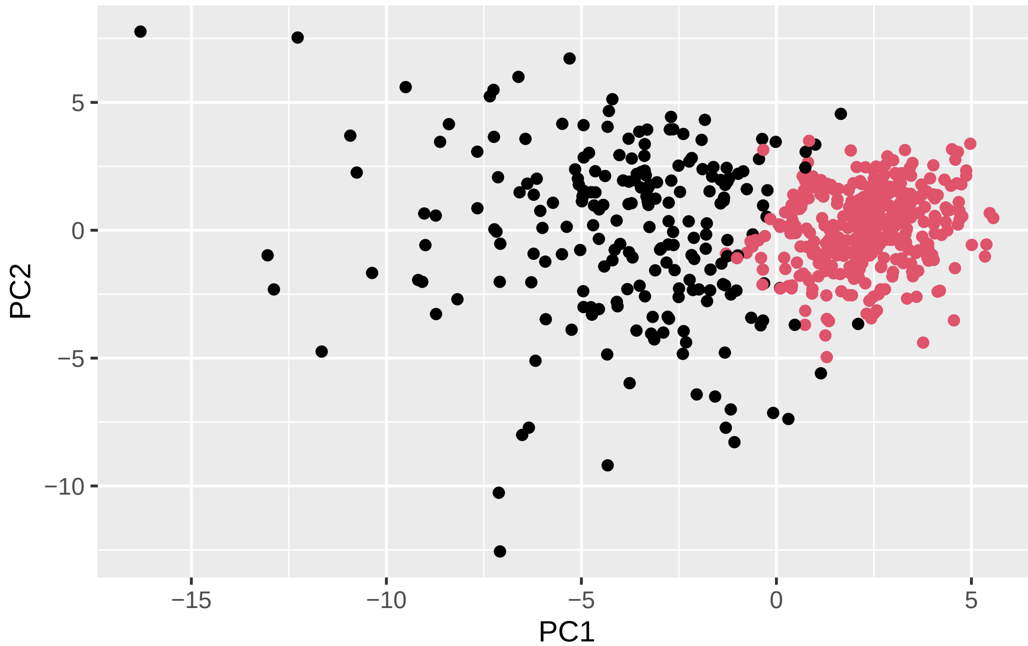

```
grps <- cutree(wisc.pr.hclust, k = 2)
table(grps)
```

```
grps
  1   2
216 353
```

```
table(grps, diagnosis)
```

```
    diagnosis
grps   B   M
   1  28 188
   2 329  24
```

```
ggplot(wisc.pr$x) +
  aes(PC1, PC2) +
  geom_point(col = grps)
```




```
# Use distance along first 7PCs for clustering
wisc.pr.hclust <- hclust(dist(wisc.pr$x[, 1:7]), method = "ward.D2")

# Cut into 2 clusters
wisc.pr.hclust.clusters <- cutree(wisc.pr.hclust, k = 2)
```

> Q13. How well does the newly created `hclust` model with two clusters separate out the two "M" and "B" diagnoses?

It separates out the two "M" and "B" diagnoses pretty well

```
table(cutree(wisc.pr.hclust, k = 2) , diagnosis)
```

```
   diagnosis
      B   M
  1  28 188
  2 329  24
```

> Q14. How well do the hierarchical clustering models you created in the previous sections (i.e. without first doing PCA) do in terms of separating the diagnoses? Again, use the `table()` function to compare the output of each model (`wisc.hclust.clusters` and `wisc.pr.hclust.clusters`) with the vector containing the actual diagnoses.

Compared to the PCA-based clustering, the hierarchical clustering models separate the the diagnoses okay. The PCA-based clustering shows a cleaner separation and separates the diagnoses better.

```
table(wisc.hclust.clusters, diagnosis)
```

```
                    diagnosis
wisc.hclust.clusters   B   M
                   1  12 165
                   2   2   5
                   3 343  40
                   4   0   2
```

```
table(wisc.pr.hclust.clusters, diagnosis)
```

```
                       diagnosis
wisc.pr.hclust.clusters   B   M
                      1  28 188
                      2 329  24
```

Prediction

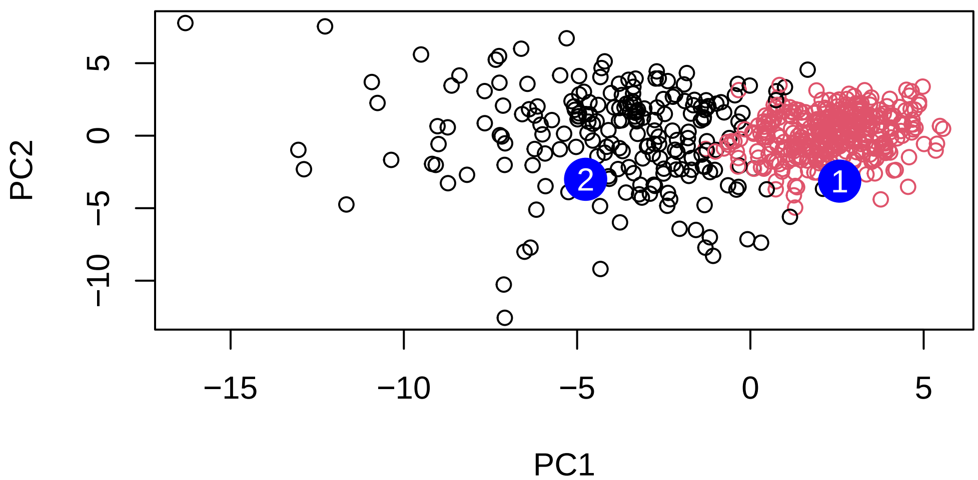
We will use the `predict()` function that will take our PCA model from before and new cancer cell data and project that data onto our PCA space.

```
url <- "new_samples.csv"
new <- read.csv(url)
npc <-predict(wisc.pr, newdata = new)
npc
```

```
           PC1       PC2        PC3        PC4       PC5        PC6        PC7
[1,]  2.576616 -3.135913  1.3990492 -0.7631950  2.781648 -0.8150185 -0.3959098
[2,] -4.754928 -3.009033 -0.1660946 -0.6052952 -1.140698 -1.2189945  0.8193031
            PC8       PC9       PC10      PC11      PC12      PC13     PC14
[1,] -0.2307350 0.1029569 -0.9272861 0.3411457  0.375921 0.1610764 1.187882
[2,] -0.3307423 0.5281896 -0.4855301 0.7173233 -1.185917 0.5893856 0.303029
          PC15       PC16        PC17        PC18        PC19       PC20
[1,] 0.3216974 -0.1743616 -0.07875393 -0.11207028 -0.08802955 -0.2495216
[2,] 0.1299153  0.1448061 -0.40509706  0.06565549  0.25591230 -0.4289500
           PC21       PC22       PC23       PC24        PC25         PC26
```

```
[1,]  0.1228233 0.09358453 0.08347651  0.1223396  0.02124121  0.078884581
[2,] -0.1224776 0.01732146 0.06316631 -0.2338618 -0.20755948 -0.009833238
             PC27        PC28         PC29         PC30
[1,]  0.220199544 -0.02946023 -0.015620933  0.005269029
[2,] -0.001134152  0.09638361  0.002795349 -0.019015820
```

```
plot(wisc.pr$x[,1:2], col=grps)
points(npc[,1], npc[,2], col="blue", pch = 16, cex = 3)
text(npc[,1], npc[,2], c(1,2), col = "white")
```



Q16. Which of these new patients should we prioritize for follow up based on your results?

Based on my results, we should prioritize Patient 2 for follow up. Their clusters are closer to the malignant cancer samples compared to Patient 1.